Data Preprocessing & Exploratory Data Analysis (EDA)

Data Mining: Earnings Surprise Prediction

EDA Report

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Project Overview and Dataset

This project investigates the idea of whether a company's four preceding recorded earnings events can help predict if its next reported earnings per share (EPS) will exceed analyst expectations. I used the *Sov.ai Earnings Surprise* dataset from Hugging Face [1]. The outcomes are classified as either an earnings 'beat' or 'meet or miss'. An earnings beat occurs when actual EPS exceeds the analyst estimate. Results equal to or below the estimate are classified as meet or miss.

The raw dataset contains 1,657,661 records spanning 5,739 tickers. After examining the repeated records and consolidating them using a provisional earnings event identifier I was able to obtain 111,989 candidate events. From these I was able to construct 90,349 eligible prediction samples with 25 historical features.

For the exploratory analysis, I decided to focus on prediction samples dated before 2024. The included Jupyter notebook contains the full data preparation process and visualizations, as well as further explanations of the methods used.

Data Quality and Preparation

I began by examining missing values and duplicate records in the raw dataset. There was no exact duplicate rows but 111,588 candidate earnings events appeared in multiple records. These observations were grouped together using the ticker and publication date, and were checked for conflicting financial values before consolidating them. The reconstruction produced 111,988 events with complete financial information and only one event was missing an actual EPS value.

The date fields raised a concern as I found several records dated before an earnings announcement that already contained the EPS eventually reported for that event. In response to this I chose to use *date_pub* to order the reconstructed events but I still cannot confirm exactly when each financial value became available.

I examined extreme financial values using the interquartile range (IQR) method. Approximately 12% to 19% of observations were flagged depending on the feature. My idea behind retaining these values is because unusually large earnings results may still be valid.

After constructing the eligible prediction samples none of the 25 historical features contained missing values. The data-quality checks and reconstruction process are explained in more detail in the notebook.

Exploratory Data Analysis

I analyzed the distributions of different historical earnings features using histograms and box plots. As shown in Figure 1, the most recent EPS surprise is concentrated near zero with fewer observations extending further in either direction. I also found that there was noticeable differences between the mean and median of certain EPS measurements which is reflective of the influence of extreme values.

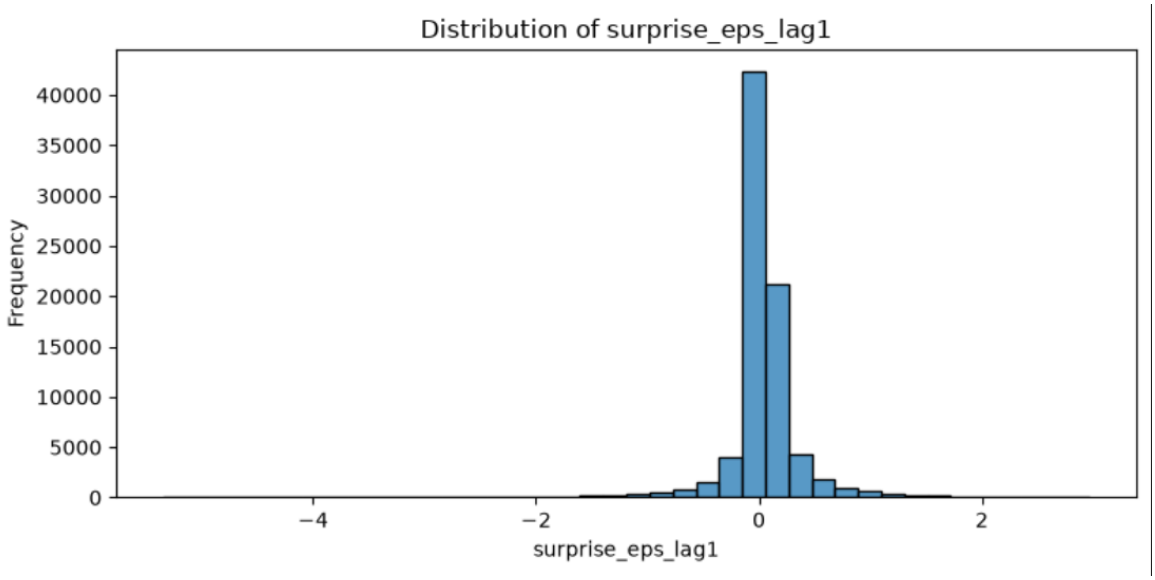


Figure 1. Distribution of the most recent historical EPS surprise showing values between the 1st and 99th percentiles.

The prediction dataset showed some class imbalance as approximately 61.5% of observations were classified as earnings beats. This is a useful fact for my later classification experiments due to the fact that a model consistently predicting an earnings beat could achieve reasonable accuracy without learning much from the historical features.

I used a Pearson correlation heatmap to examine relationships between the numerical predictors. As can be seen in Figure 2 the most recent actual EPS and earnings surprise had a strong positive correlation of 0.97. I partly expected this relationship due to the fact that earnings surprise is calculated using actual and estimated EPS. The extreme values that were identified earlier may have also had an impact on the correlations which is worth considering when interpreting these features.

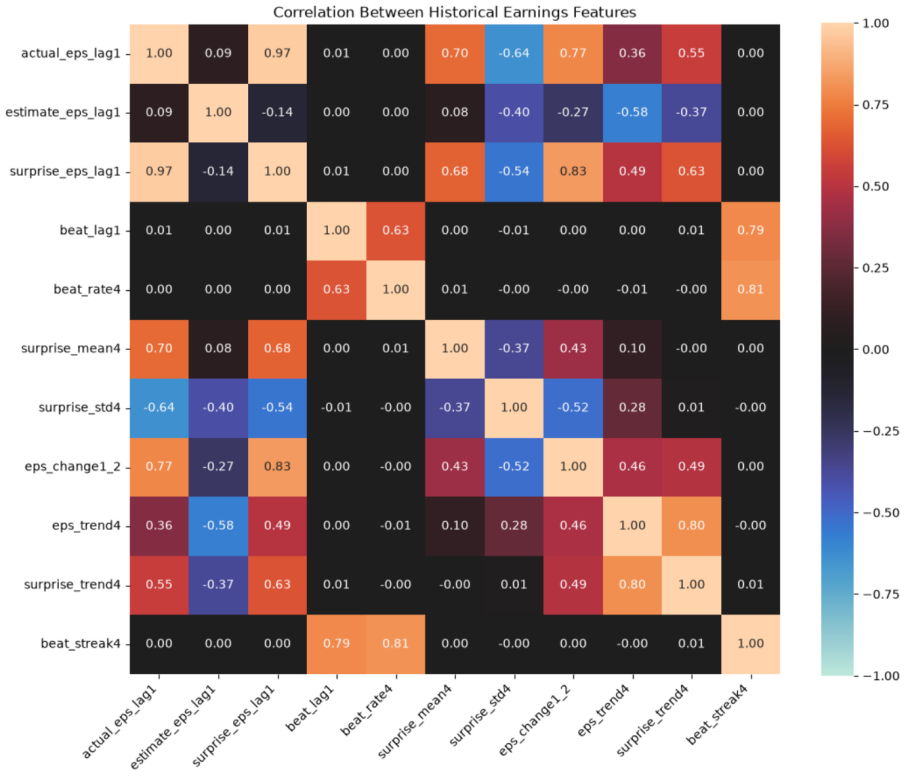


Figure 2. Pearson correlation matrix for selected historical earnings features in the development dataset.

Preprocessing Methods and Findings

I divided the development data chronologically using observations before 2023 for training and those from 2023 for validation. All preprocessing transformations were fitted using the training data to avoid information leakage. The eligible prediction samples contained no missing feature values, though I did include imputation in the pipeline to handle incomplete observations in later experiments.

Given the extreme earnings values identified during EDA, I used *RobustScaler* to scale continuous features using the median and interquartile range. The categorical month and quarter variables were converted using one-hot encoding.

I used mutual information to rank the transformed features based on their relationship with the earnings-beat outcome. For my initial analysis I kept the 15 highest-scoring columns. Historical beat frequency received the highest score at 0.0464 and beat streak length got a score of 0.0397. I found these rankings to be interesting because both features describe patterns across previous earnings announcements. I intend on further investigating whether keeping this smaller set improves classification results when I begin comparing the models.

I applied PCA to examine the dimensionality of the preprocessed features. The first two components explained 86.98% of the variance but this decreased to 66.72% when extreme values were capped at the 1st and 99th percentiles which further reinforced my concern about the influence of unusually large financial observations.

I did not apply data augmentation or class resampling because the dataset already contains over 90,000 eligible samples with a moderate class imbalance. I provided further explanations and visualizations of these preprocessing decisions in the accompanying notebook.

References

[1] Sov.ai, "Earnings Surprise," Hugging Face, [Online]. Available: https://huggingface.co/datasets/sovai/earnings_surprise. [Accessed 3 October 2026].